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INTRODUCTION: CCHS

- CCHS (Congenital Central Hypoventilation Syndrome)** is a rare autonomic disorder that impairs breathing regulation (Figure 1), necessitating **lifelong artificial respiratory support** [1].
- In ambulatory children with CCHS, bilateral transthoracically-implanted **phrenic nerve pacers** (Avery Biomedical Devices; Commack, NY, USA) provide artificial daytime ventilation by electrically stimulating the phrenic nerves to cause diaphragm contractions [2] (Figure 2).

DESIGN NEED

- To guide pacer management in pediatric patients, clinicians monitor **exertion level and respiratory demand via motion sensing and end-tidal CO₂ (EtCO₂)**.
- Current methods rely on bulky oximetry-capnography systems and wearable motion-tracking devices (Hexoskin; Montréal, QC, Canada), **limiting assessments to in-hospital clinical environments**.

We developed a portable device that synchronously records **EtCO₂** and **six degrees of freedom (6-DoF) motion data** to enable **continuous, non-invasive respiratory monitoring** during activities of daily living (Figures 3 and 4).

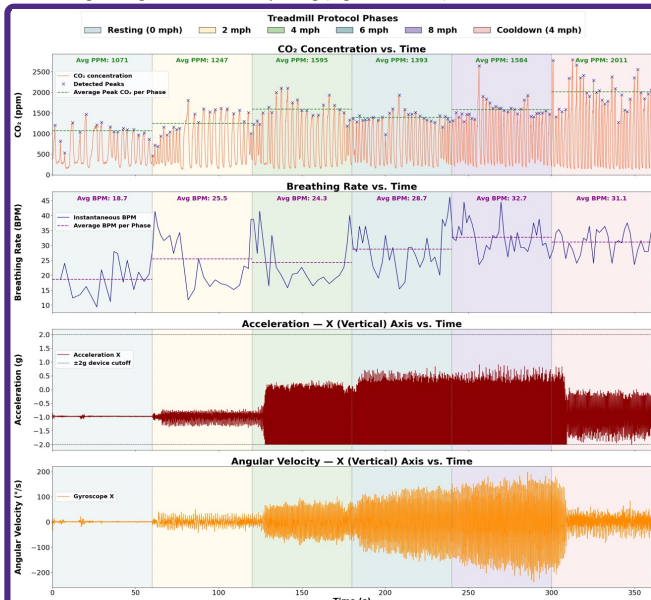


Figure 5. Representative data from treadmill trial experiments using the modified BCCH protocol. Raw EtCO₂ and 6-DoF IMU data were plotted across six treadmill phases. In this trial, a 21-year-old male participant reached volitional fatigue at 5 minutes, followed by a 1-minute cooldown.

Note: The CO₂ signal is uncalibrated. Accelerometer output was limited to $\pm 2g$ due to the IMU's default configuration. Upcoming work will focus on optimizing both IMU and CO₂ sensor settings.

FRONT VIEW



BACK VIEW

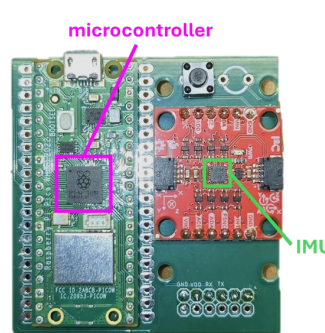


Figure 3. The device includes a custom-designed printed circuit board (PCB) which includes a microcontroller, a 6-DoF IMU (100 Hz), a non-dispersive infrared CO₂ sensor (20 Hz), and a microSD card reader for data storage. The PCB is enclosed in a 3D-printed case and secured to the user's chest during data collection. Synchronous EtCO₂ and motion data are stored in a CSV format with C-based firmware for memory optimization and power efficiency. Additional features include a micro-USB programming port, a rechargeable Li-ion battery, indicator LEDs (CO₂ sensor, IMU, & microSD), and start/stop buttons.

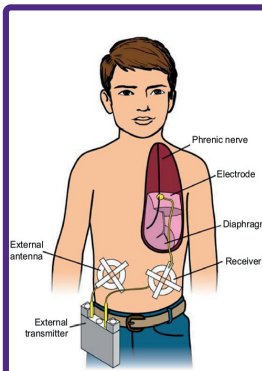


Figure 2. Setup of a surgically implanted phrenic nerve pacer in a pediatric patient with CCHS [3]. Phrenic nerve pacers support ventilation during physical activity by delivering electrical pulse trains to the phrenic nerve, triggering diaphragm contractions and lung expansion. Parameters such as pulse width, amplitude, interval, range, and slope can be adjusted. However, current pacers operate as "open-loop" systems, relying entirely on caregiver input to set pacing parameters across varying activity levels.

CCHS During Exercise

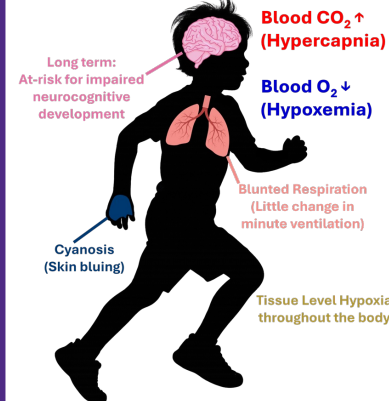


Figure 1. CCHS is a rare disorder of the autonomic nervous system that impairs the body's ability to regulate breathing. Affected individuals exhibit blunted respiratory responses, characterized by a diminished ability to adjust ventilation in response to physiological stimuli such as elevated blood carbon dioxide (CO₂) levels (hypercapnia) and reduced oxygen (O₂) levels (hypoxemia).

DEVICE TESTING: TREADMILL EXPERIMENT

- Modified version of the **British Columbia Children's Hospital (BCCH) treadmill protocol** for pediatric patients [4].
 - An initial rest phase of 1 minute to assess baseline respiration.
 - Treadmill set to 2.0 miles per hour (mph) at a constant 1% incline.
 - Speed increased by 2.0 mph every minute until test participant reached volitional fatigue, followed by a 1-minute cooldown at 4 mph.
- Conclusion:** EtCO₂ and motion data were analyzed to extract respiratory and motion features (Figure 5), demonstrating the device's utility as a portable tool for clinically relevant data collection.

FUTURE DIRECTIONS: AUTONOMOUS PACING

- Current pacers use open-loop, manually-set parameters, potentially leading to **suboptimal ventilation** during physical activity and **markedly increased caregiver burden** [2].
- We aim to train a model that predicts EtCO₂ from motion data, enabling **real-time estimation of respiratory demand to autonomously adjust pacer settings**.
- Future work involves large-scale studies in healthy children to develop a "digitally implantable" biofeedback system that **mimics normal autonomic control for children with CCHS**.

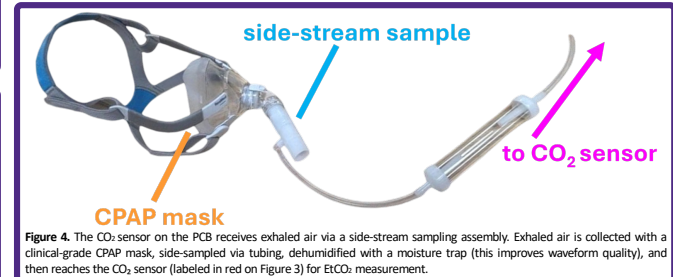


Figure 4. The CO₂ sensor on the PCB receives exhaled air via a side-stream sampling assembly. Exhaled air is collected with a clinical-grade CPAP mask, side-sampled via tubing, dehumidified with a moisture trap (this improves waveform quality), and then reaches the CO₂ sensor (labeled in red on Figure 3) for EtCO₂ measurement.

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